

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV - Semester of BSc (Biochemistry) Examination April 2019

Course code & Name BC 406: Enzymology

Date: 03-05-2019 Day: Friday Time: 1:30 PM To 2:00 PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	The scientist who coined the term “enzymes” is.....		
	(i) F.W. Kuhne	(ii) Edward Buchner	
	(iii) Louis Pasteur	(iv) James Sumner	
2.	During _____ the final product of a metabolic pathway turn off the first step of metabolic pathway.		
	(i) Positive feedback	(ii) Negative feedback	
	(iii) Competitive feedback	(iv) Both A and C	
3.	Pepsin is an example for.....		
	(i) Proteolytic enzyme	(ii) Zymogen	
	(iii) Hydrolases	(iv) All	
4.	Which of the following enzymes are widely used in detergent industry?		
	(i) Papain	(ii) Celluloses	
	(iii) Pectinases	(iv) Lipases	

5.	The activities of many enzymes, membrane transporters and other proteins can be quickly activated or inactivated by phosphorylation of specific amino acids residues. This regulation is called		
	(i) Allosteric modification	(ii) Covalent modification	
	(iii) Irreversible modification	(iv) Feedback regulation	
6.	Which of the following is not a cofactor?		
	(i) Co-enzyme	(ii) Apoenzyme	
	(iii) Prosthetic group	(iv) Metal ion	
7.	Which statement is <u>incorrect</u> for enzyme catalyst reaction		
	(i) Reduction in activation energy		
	(ii) No change in overall free net energy		
	(iii) Shift the position of chemical equilibrium		
	(iv) Attain the chemical equilibrium faster		
8.	Which of the following protein purification method is based on polarity?		
	(i) Adsorption chromatography	(ii) Paper chromatography	
	(iii) Hydrophobic interaction chromatography	(iv) All of the above	
9.	For preparing a protein calibration graph, which of the following Albumin is used as a standard protein.		
	(i) Salmon sperm	(ii) Calf thymus	
	(iii) Bovine serum	(iv) Ribozyme	
10.	All of the following statements are true about isozymes except:		
	(i) They are different form of the enzyme that catalyze the same reaction		
	(ii) They use the same substrate and produce the same product		
	(iii) They have the same physical properties		
	(iv) They are found in different tissues		
11.	Enzymatic conversion of Sucrose to glucose and fructose can be estimated by...		
	(i) Calorimeter	(ii) U.V spectrophotometer	
	(iii) Fluorometer	(iv) Polarimeter	
12.	Michaelis and Menten assumed that the overall reaction for an enzyme-catalyzed could be written as: $E + S \xrightleftharpoons[K_{-1}]{K_1} ES \xrightarrow{K_2} P$ Using this reaction, the rate of breakdown of the enzyme-substrate complex can be described by the expression:		
	(i) $K_1 ([Et] - [ES])[S]$	(ii) $K_{-1} [ES] + K_2 [ES]$	
	(iii) $K_2 [ES]$	(iv) $K_{-1} [ES] + K_1 [ES]$	

13.	Which of the following scientist postulated that “All enzymes are proteins?”		
	(i) Louis Pasteur	(ii) James Sumner	
	(iii) Edward Buchner	(iv) D. Koashland	
14.	The first step in any reaction catalysed by an enzyme is the formation of a specific association between the molecules called an.....		
	(i) Enzyme-product complex	(ii) Enzyme-intermediate complex	
	(iii) Enzyme-substrate complex	(iv) None of these	
15.	Which of the following enzyme is an example for group specificity for oxidation of various alcohols...?		
	(i) Glucokinase	(ii) Hexokinase	
	(iii) Lactate dehydrogenase	(iv) Alcohol dehydrogenase	
16.	Which of the following salt is most widely used for precipitating the enzymes...?		
	(i) Ammonium sulphate	(ii) Ammonium per sulphate	
	(iii) Ammonium chloride	(iv) Ammonium per chloride	
17.	When $[S] = 0.1 \cdot K_M$, the velocity of an enzyme catalyzed reaction is about:		
	(i) $0.1 \cdot V_{\max}$	(ii) $0.3 \cdot V_{\max}$	
	(iii) $0.5 \cdot V_{\max}$	(iv) $0.7 \cdot V_{\max}$	
18.	Which of the following is an example for gastric zymogen...?		
	(i) Trypsinogen	(ii) Chymotrypsinogen	
	(iii) Pepsinogen	(iv) All	
19.	Enzymes frequently used in ELISA are.....		
	(i) Isomerases	(ii) Lysozymes	
	(iii) Ligases	(iv) None	
20.	Which of the following statements is correct about allosteric enzyme?		
	(i) Allosteric enzymes are usually monomeric enzyme		
	(ii) Allosteric enzymes follow Michaelis –Menten kinetics		
	(iii) Allosteric enzymes show reduced sensitivity to changes in substrate concentration compared to classical type enzymes with hyperbolic kinetics.		
	(iv) Allosteric enzymes show greater sensitivity to changes in substrate concentration compared to classical type enzymes with hyperbolic kinetics.		